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General Education Reflection

My major classes in computer engineering teach me how to design and construct a system, and my general education classes teach me about the people and situations that the system will impact. So far, I have taken ECON 101, ECON 102, RUS 375, and INTST 235. These courses have given me a new and improved way of looking at engineering problems in an economic, cultural and global perspective. I haven't finished my U.S. Diversity requirement.

I studied scarcity, opportunity cost, and supply and demand in ECON 101. These ideas relate to engineering in that all projects have finite resources and competing priorities. Engineers are faced with a dilemma when they have to select components, or design software, as they must also take into account cost as well as performance and reliability. The opportunity cost is a good tool to help consider these decisions: If time or money is invested in one feature, then fewer resources are available for another feature. This becomes especially important for the digital divide, where a technologically successful product might not be used if individuals who need it are unable to pay for it.

ECON 102 has given me an understanding of the bigger picture of economics, such as inflation, unemployment, and economic growth. These ideas demonstrate the need to consider factors outside of the design when making engineering decisions. Price fluctuation and resource

availability can impact project costs, manufacturing costs and feasibility of making a product. For instance, when the semiconductors fail to flow, engineers must think of other components. Engineers need to be aware of the economic context to make informed decisions that consider both technical needs and practical considerations.

Through RUS 375, I was given a chance to look at other cultures' points of view. When I studied another society, it really made me think twice about how people talk, how people think they interpret information, how people think they interpret it, how people react to things, what they do in different situations. This is important for engineering because people are different and have different expectations of what they are using. What is easy to understand to the developer might be hard to understand by other users. This is why it is important to take cultural differences into account to enable engineers to communicate effectively and create products that will benefit more clients.

INTST 235 gave me the opportunity to consider problems with a transnational perspective. Engineering projects frequently have people working from different countries, suppliers and users, so what decisions are made in one place can impact on another location. The sustainability of the environment is one. Engineers must also consider energy consumption, use of materials, and electronic wastes in addition to functionality when designing electronic systems. A global view makes me realize that a product can have ramifications beyond the use of the initial consumer.

These courses have broadened my thinking and approach to solving engineering problems. I must ask myself if a solution is affordable, understandable and responsible and if it

works. This will be a chance to revisit considerations for engineering based on varying experiences and needs as the U.S. Diversity requirement.